

Low Power Design

ISIP LAB
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Power Intro.

1. Reason for low power:

- **Power**↑ → **Temperature**↑ → **Safety**↓
- May lead to permanent damage to the chip
- Portable device require low power

2. $P_{\text{total}} = P_{\text{static}} + P_{\text{dynamic}}$

Static Power

$$P_{\text{static}} = I_{\text{leakage}} * V_{\text{dd}}$$

- Reverse leakage current
 - Happen when device is reverse biased
 - Increase with temperature
- Sub-threshold current
 - Dominate the leakage current
 - increase with temperature
- Gate Leakage current
 - Increase with V_{dd}

Dynamic Power

$$P_{\text{dynamic}} = p_t * (P_{\text{sw}} + P_{\text{sc}})$$

- p_t : Switching Probability
- P_{sw} : Switching Power
 - $P_{\text{sw}} = C_L * V_{\text{dd}}^2 * f_p$
 - Charging and discharging the loading capacitive
- P_{sc} : Short circuit power
 - Both NMOS and PMOS are turned on

How to minimize power

- Static:
 - Reduce voltage supply (Process dependent)
 - In general: the leakage current can't be changed if process and cells are decided
- Dynamic :
 - Reduce Vdd (Process dependent) P_{sw} & P_{sc} (e.g. Multi-Voltage)
 - Reduce parasitic capacitance P_{sw}
 - Reduce the overlap time of PMOS and NMOS turn-on time P_{sc}
 - Reduce unnecessary switching activities p_t (e.g. Clock Gating)

Thank